# GAM Lab User Study

Interactive review and refinement of an explainable AI model for predicting the probability of mortality among intensive care patients.

# Session Procedure

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| PART1 | Pre-Study Interview — Professional Background & Practice |
| PART2 | System Introduction — MIMIC-IV, GAM, Interface Tour |
| PART3 | Main Task — Plausibility Check |
| PART4 | Post-Study Interview |

PART 1

# Pre-Study Interview

Professional background, the role of prognoses in everyday clinical practice, and working with IT systems—four brief topic blocks.

# Professional Background

01 Can you briefly summarize your professional experience in the medical field?

02 Do you have any specific experience in the field of critical care medicine?

# Prognoses in Everyday Work

01 What role do patient prognoses or clinical assessments play in your daily work?

02 How often do you make prognoses yourself or are involved in their development?

03 What types of prognoses or assessments are typically involved?

# Personal Involvement

01 Are you personally involved in such assessments or decisions?

# Digital Tools

01 Do you use IT systems when making prognoses or clinical assessments?

02 Which IT systems or digital tools do you use for this purpose?

PART 2

# System Introduction

Explanation of the setup, the dataset, and the actual model—followed by a brief tour of the interface.

# MIMIC-IV Dataset

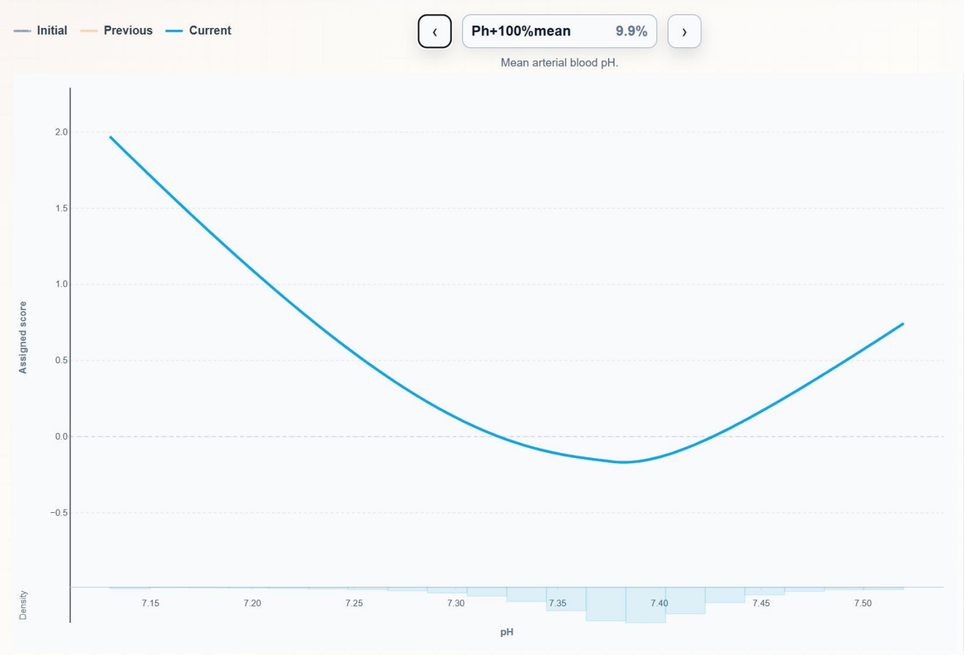
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|---|---|
| SOURCE | Routine clinical data, Beth Israel Deaconess Medical Center, Boston (2008–2022). |
| STAY DATA | Data documented during the stay in the intensive care unit |
| USAGE | 10 variables—vital signs and laboratory values. **Mean over the first 24 hours of the ICU stay.** |
| OUTCOME VARIABLE | Does the patient survive the stay in the intensive care unit—yes or no. |

# What is a GAM?

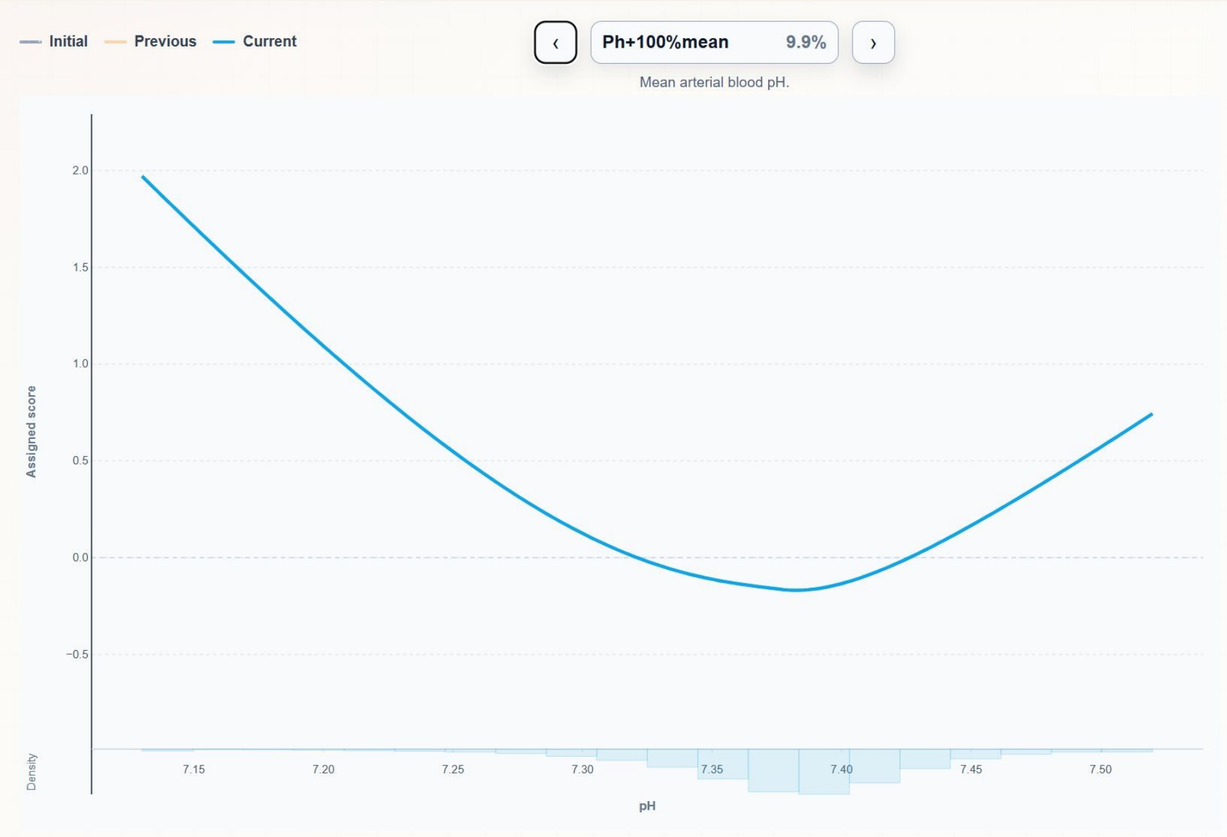
An interpretable statistical prediction model. The overall prediction is derived by **summing the contributions of individual variables**—e.g., age, heart rate, respiratory rate.

For each variable, there is a **function** that shows how a measured value influences the risk score.

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| X-AXIS | Measured value of the variable—e.g., heart rate in beats per minute |
| Y-AXIS | Risk Score — 0 neutral, + higher, − lower |



# Example Shape Curve



# Interface Tour

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| COMPONENTS | Walk through the key elements of the interface together. |
| FUNCTIONS | Get familiar with the available functions and information. |
| EXERCISE | Try out selected functions using short examples. |

PART 3

# Main Task

Check the plausibility of all the model’s curves—think aloud, go through them step by step, and make adjustments.

# Plausibility check of the curves

The variables represent patient measurements—averaged over the first 24 hours of their stay in the intensive care unit.

Please go through all the functions one by one and describe whether the trends make sense based on your clinical experience. Say aloud what stands out to you—an unexpected trend, a suspicious kink, or a range that doesn’t seem right.
If possible, make adjustments to the functions.

PART 4

# Post-Study Interview

Four topic areas—ranging from overall impression to comprehensibility and trust, to the requirements for a trustworthy system.

# Overall Impression

01 How would you describe your overall impression of the system after using it?

02 Was there anything that pleasantly surprised you?

03 Was there anything that confused or unsettled you?

# Comprehensibility

01 How well were you able to understand what the graphs represented?

02 Which parts of the presentation were particularly helpful? Were there any elements that were difficult to understand or ambiguous?

03 How did you understand the presentation of the individual variable contributions to the probability of death?

# Trust

01 How did using the tool influence your assessment of the model?

02 Did the tool generate trust, skepticism, or a nuanced assessment?

03 Under what conditions would you trust—or not trust—such a model?

04 Did you feel like you could control the model?

# Requirements

01 What additional information would be important?

02 What requirements would such a system need to meet for you to find it useful in a clinical context?

03 What would need to be displayed—and what monitoring capabilities would you need?

# Thank you very much.

Thank you for your time, your candor, and your clinical assessment.